

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
In [2]: data = pd.read_csv("housing.csv")
data.head()
```

```
Out[2]:
```

	ID	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat	medv
0	1	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	4.98	24
1	2	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	9.14	21
2	4	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2.94	33
3	5	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	5.33	36
4	7	0.08829	12.5	7.87	0	0.524	6.012	66.6	5.5605	5	311	15.2	395.60	12.43	22

```
In [5]: X = data.drop("medv", axis=1)
y = data["medv"]
```

```
In [6]: X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
In [8]: model = LinearRegression()
```

```
In [9]: model.fit(X_train, y_train)
```

```
Out[9]:
```

▼ LinearRegression ⓘ ?
 LinearRegression()

```
In [10]: y_pred = model.predict(X_test)
```

```
In [16]: y_pred
print(y_pred)
```

```
[25.2066139 23.01556619 23.02881703 32.86920564 24.8010857 14.11165892
17.47866789 29.93450929 15.56339277 25.10055834 26.5902006 19.9993111
20.16486173 34.27598213 21.98742925 34.45916224 22.37932546 16.49547941
25.51139099 16.99443916 36.4075702 31.40322718 22.97022721 27.92712854
16.55701206 42.27567357 7.52094503 -0.06547566 30.97901131 9.28545638
19.09797694 20.59200958 27.44242591 14.50456562 19.01730989 12.12396659
27.61602987 4.38411748 17.0819314 22.51905104 24.11941368 21.82263621
25.2651197 39.1228954 36.32182138 21.38932664 10.77563809 21.46350357
14.02722307 20.32510853 13.18431501 28.21198001 21.71122344 13.40409046]
```

```
33.99377333 0.88546196 20.3995266 27.0965807 25.1792521 28.43350073
17.94360554 24.04426838 18.10206696 28.12607847 23.55285523 16.69631901
26.40329812]
```

```
In [11]: mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Mean Squared Error (MSE):", mse)
print("R-squared (R²) value:", r2)
```

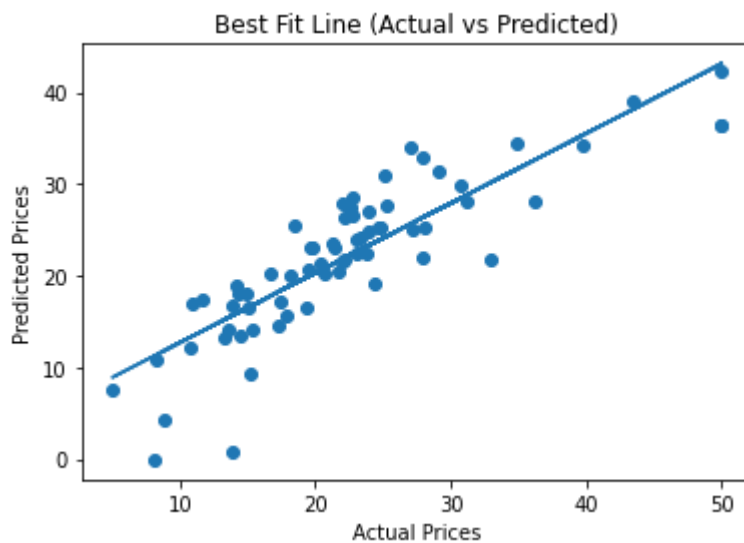
```
Mean Squared Error (MSE): 23.41170182659855
R-squared (R²) value: 0.7398653051224333
```

```
In [14]: import matplotlib.pyplot as plt
import numpy as np

plt.scatter(y_test, y_pred)

m, b = np.polyfit(y_test, y_pred, 1)
plt.plot(y_test, m*y_test + b)

plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")
plt.title("Best Fit Line (Actual vs Predicted)")
plt.show()
```



```
In [18]: X = data[["rm"]]
y = data["medv"]
```

```
In [19]: X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)
```

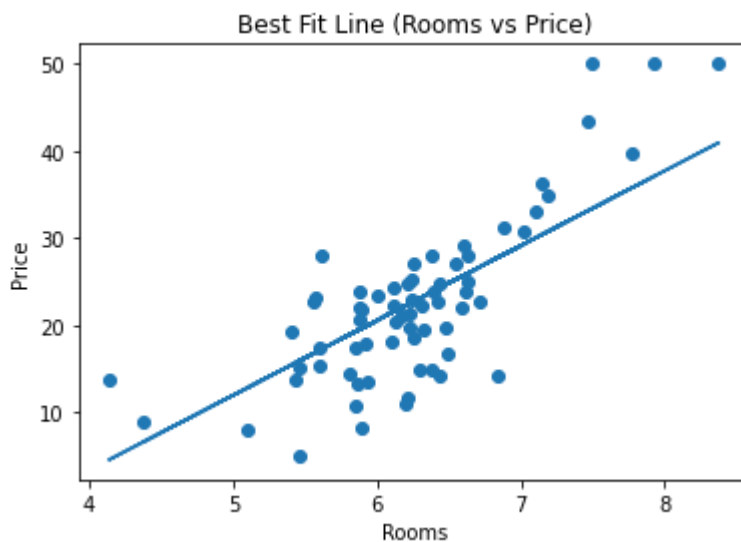
```
Out[19]: LinearRegression
LinearRegression()
```

```
In [20]: y_pred = model.predict(X_test)
```

```
In [21]: print("MSE:", mean_squared_error(y_test, y_pred))
print("R2:", r2_score(y_test, y_pred))
```

MSE: 36.361622515889756
R2: 0.5959747117709422

```
In [22]: plt.scatter(X_test, y_test)
plt.plot(X_test, y_pred)
plt.xlabel("Rooms")
plt.ylabel("Price")
plt.title("Best Fit Line (Rooms vs Price)")
plt.show()
```



```
In [24]: #check the liner variable with different variables, compare where improving a
```

```
In [25]: X = data[["lstat"]]
y = data["medv"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, rand

model = LinearRegression()
model.fit(X_train, y_train)

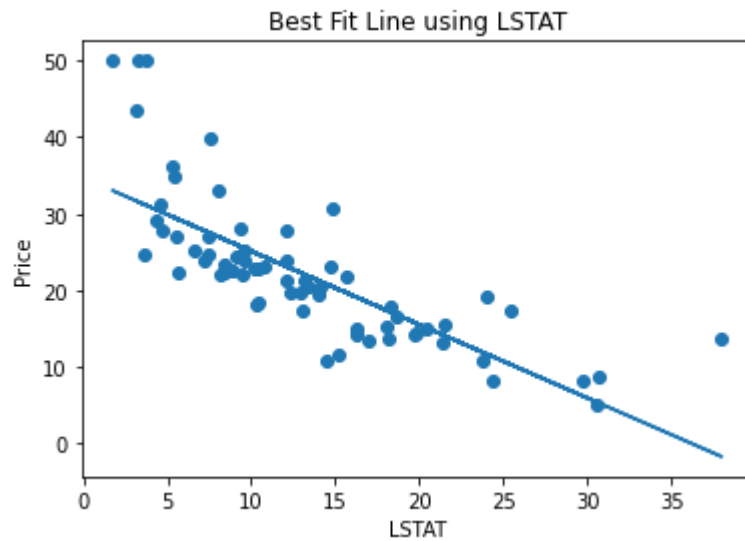
y_pred = model.predict(X_test)

print("LSTAT Results")
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2:", r2_score(y_test, y_pred))

plt.scatter(X_test, y_test)
plt.plot(X_test, y_pred)
plt.xlabel("LSTAT")
plt.ylabel("Price")
plt.title("Best Fit Line using LSTAT")
plt.show()
```

LSTAT Results
MSE: 37.73764959374321

R2: 0.5806852472675971



In [28]:

```

X = data[["age"]]
y = data["medv"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, rand

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("AGE Results")
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2:", r2_score(y_test, y_pred))

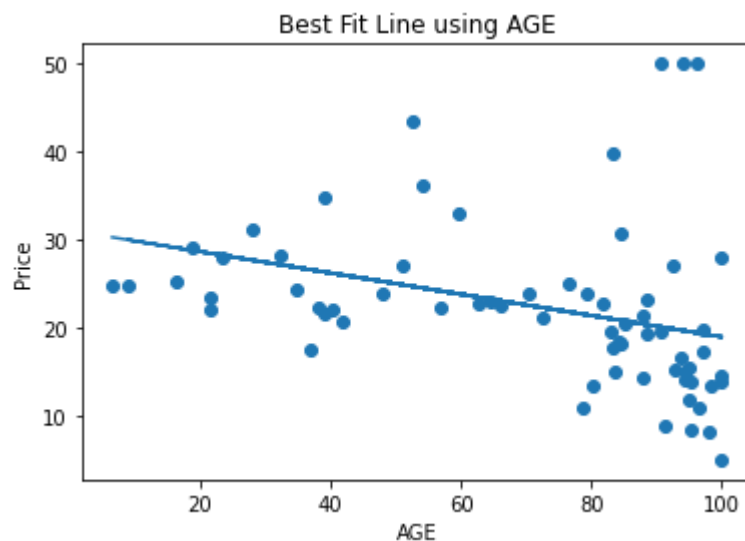
plt.scatter(X_test, y_test)
plt.plot(X_test, y_pred)
plt.xlabel("AGE")
plt.ylabel("Price")
plt.title("Best Fit Line using AGE")
plt.show()

```

AGE Results

MSE: 82.85131344011474

R2: 0.07941330785856782



In [29]:

```
X = data[["tax"]]
y = data["medv"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, rand

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

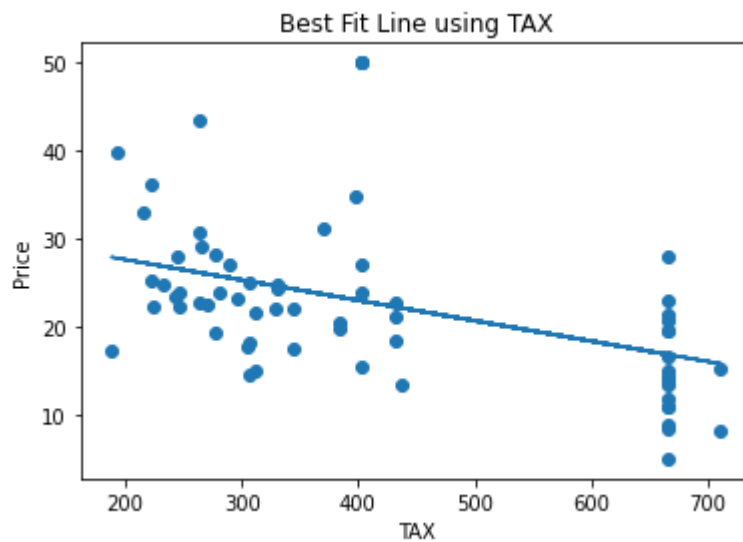
print("TAX Results")
print("MSE:", mean_squared_error(y_test, y_pred))
print("R2:", r2_score(y_test, y_pred))

plt.scatter(X_test, y_test)
plt.plot(X_test, y_pred)
plt.xlabel("TAX")
plt.ylabel("Price")
plt.title("Best Fit Line using TAX")
plt.show()
```

TAX Results

MSE: 67.4736756565273

R2: 0.2502790203300326



In []: